THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 6 & 10: **Triangles, Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 8 cm	$\frac{3}{4} = \frac{6}{EC}$
2	**(a)** $\frac{2}{3}$	Ratio of corresponding sides
3	**(a)** AAA	Equiangular triangles
4	**(c)** Incorrect — their angles need not be equal	e.g. $70°, 70°, 40°$ vs $50°, 50°, 80°$
5	**(b)** Student A is correct	$\frac{2}{3} = \frac{h}{12}$
6	**(b)** 4 cm	$\sqrt{25-9}$
7	**(d)** $130°$	Supplementary
8	**(a)** 2	One on each side
9	**(a)** Incorrect — it is perpendicular to the radius through that point	Theorem 10.1
10	**(a)** Student A is correct	$2\sqrt{25-9}$; B gave half the chord

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{2.4}{DB} = \frac{3.2}{4.8} \Rightarrow DB = 3.6$ cm	1+1
12	$AB = 6$ cm, $BC = 8$ cm	1+1
13	$\angle AOB = 110°$; $\triangle OAB$ isosceles $\Rightarrow \angle OAB = 35°$	1+1
14	$AB + CD = BC + AD \Rightarrow AD = 8$ cm	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	BPT: $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{DB}{AD} = \frac{EC}{AE} \Rightarrow$ add 1: $\frac{AB}{AD} = \frac{AC}{AE} \Rightarrow$ result	1+1+1
16	Lines touch at P, Q; $OP = OQ$, $\angle OPA = \angle OQA = 90°$, OA common $\Rightarrow \triangle OPA \cong \triangle OQA$ (RHS) $\Rightarrow \angle OAP = \angle OAQ$	1+1+1
17	$\frac{PQ}{BC} = \frac{AP}{AB} = \frac{1}{3} \Rightarrow PQ = 3$ cm; $\frac{AQ}{QC} = \frac{AP}{PB} = \frac{1}{2}$	2+1
	OR: $\triangle AOB \sim \triangle COD \Rightarrow \frac{OA}{OC} = \frac{AB}{CD} = 2$; $OA = 6$, $AC = 9$ cm	2+1
18	$r = \sqrt{169-144} = 5$ cm; nearest distance $= 13 - 5 = 8$ cm	2+1
	OR: $\triangle OPT \cong \triangle OQT$ (RHS: $OP = OQ$, OT common, right angles) $\Rightarrow \angle OTP = \angle OTQ$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\angle ABD = \angle ACB = \angle ECF$, $\angle ADB = \angle EFC = 90° \Rightarrow$ AA (3). $\frac{AB}{EC} = \frac{AD}{EF} \Rightarrow EF = 4$ cm (2)	3+2
	OR: (i) AA (ii) $\frac{4}{2.5} = \frac{h}{15} \Rightarrow h = 24$ m (iii) $\frac{4}{s} = \frac{24}{20} \Rightarrow s = \frac{10}{3} \approx 3.33$ m	1+2+2
20	(i) $AB = 10$, $BC = 9$, $CA = 11$ cm (ii) 30 cm (iii) $15 = \frac{30}{2}$ ✓ (iv) tangents from an external point are equal	2+1+1+1
	OR: RHS congruence proof (3); $\triangle PAB$ is equilateral $\Rightarrow AB = 10$ cm (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapters 6 & 10: **Triangles, Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50°	A corresponds to Q
2	**(a)** 2 cm	$\frac{DE}{6} = \frac{1}{3}$
3	**(d)** one angle equal and the sides including it proportional	Criterion
4	**(b)** Incorrect — equal area does not imply equal angles	Area alone decides nothing
5	**(b)** Student A is correct	Ratio 1 : 2 throughout
6	**(c)** tangent	Definition
7	**(b)** 24 cm	$\sqrt{625 - 49}$
8	**(c)** 80°	$180^\circ - 100^\circ$
9	**(b)** Correct	$\triangle OPT \cong \triangle OQT$
10	**(b)** Student A is correct	$5 + 7 = 6 + AD$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{3}{6} = \frac{4}{8} = \frac{6}{12} = \frac{1}{2}$ ⇒ similar (SSS)	1+1
12	$(4x - 3)(5x - 3) = (8x - 7)(3x - 1) \Rightarrow 2x^2 - x - 1 = 0$ ⇒ $x = 1$ ($x = -\frac{1}{2}$ rejected)	1+1
13	$XC = XA$, $YC = YB$ ⇒ perimeter $= PA + PB = 20$ cm	2
14	$PT = 15$ cm; $\angle OTP = 90^\circ$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$TR = 5$ cm; $\frac{ST}{9} = \frac{3}{9} \Rightarrow ST = 3$ cm	2+1
16	Each such chord is bisected at the point of contact: length $= 2\sqrt{R^2 - r^2}$, the same for all	1+1+1
17	$AE = \frac{2}{5} \times 10 = 4$ cm; $EC = 6$ cm	1+1+1
	OR: $\frac{AO}{OC} = \frac{BO}{OD} \Rightarrow 2x^2 - 5x + 2 = 0$ ⇒ $x = 2$ ($x = \frac{1}{2}$ makes OC negative)	1+1+1
18	$TA = TB$ ⇒ $\angle TAB = 50^\circ$; $\angle OAB = 90^\circ - 50^\circ = 40^\circ$	1+1+1
	OR: $AS = 18 = AR$ ⇒ $BR = 29 - 18 = 11$; OQBR is a square ⇒ $r = 11$ cm	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\angle A = \angle F$, $\angle C = \angle G \Rightarrow \angle ACD = \angle FGH$ ⇒ $\triangle ACD \sim \triangle FGH$ (AA) ⇒ (i) and (iii); similarly with $\angle B = \angle E$ ⇒ (ii)	2+2+1
	OR: (i) AA (ii) $\frac{1.5}{2} = \frac{h}{30} \Rightarrow h = 22.5$ m (iii) $\frac{1.5}{1.2} = \frac{22.5}{s} \Rightarrow s = 18$ m	1+2+2
20	(i) $\sqrt{16900 - 14400} = 50$ m (ii) equal tangents (iii) 90° (iv) $120 + 50 + 50 + 120 = 340$ m	2+1+1+1
	OR: Shortest-distance proof (3); $OQ = \sqrt{25 + 144} = 13$ cm (2)	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 6 & 10: **Triangles, Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 7.5 m	$\frac{1.5}{2} = \frac{h}{10}$
2	**(c)** 60°	$\angle D = \angle A$
3	**(b)** 10 cm	$\frac{6}{BC} = \frac{3}{5}$
4	**(c)** Incorrect — they are similar, not necessarily congruent	AA gives similarity
5	**(a)** Student A is correct	All angles 60°
6	**(c)** 1	Definition
7	**(d)** $6\sqrt{2}$ cm	OAPB is a square; OP is its diagonal
8	**(a)** 24 cm	$\sqrt{676 - 100}$
9	**(a)** Incorrect — a secant cuts the circle at two points	Definition
10	**(a)** Student A is correct	$\angle PTQ = 2\angle OPQ$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$EC = 12$ cm ⇒ $AC = 20$ cm	1+1
12	$\angle C = 40^\circ$, $\angle P = 60^\circ$ ⇒ $\triangle ABC \sim \triangle PQR$ (AA)	1+1
13	$\sqrt{625 - 576} = 7$ cm	2
14	$\angle PTQ = 70^\circ$; $TP = TQ$ ⇒ $\angle TPQ = 55^\circ$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AFD \sim \triangle EFB$ (vertically opposite angles; $\angle DAF = \angle BEF$ alternate) ⇒ $\frac{DF}{FB} = \frac{FA}{EF}$ ⇒ result	1+1+1
16	$\triangle OPT \cong \triangle OQT$ (RHS) ⇒ $\angle POT = \angle QOT$	1+1+1
17	$h = 50$ m; then $\frac{10}{10} = \frac{50}{s} \Rightarrow s = 50$ m	2+1
	OR: $\frac{3}{6} = \frac{5}{10}$ ⇒ $PQ \parallel BC$ (converse BPT); $PQ = \frac{1}{3} \times 12 = 4$ cm	2+1
18	$2x + 9 = x + 4 + 7 \Rightarrow x = 2$; $AB = 4$, $BC = 6$ cm	1+1+1
	OR: The radii to both points of contact are perpendicular to the parallel tangents, so they lie on one line through O	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) AA (ii) $\frac{20}{AB} = \frac{30}{75} \Rightarrow AB = 50$ m (iii) $50 - 20 = 30$ m	1+2+2
	OR: Mid-point theorem: $EF = \frac{1}{2}BC$, $FD = \frac{1}{2}CA$, $DE = \frac{1}{2}AB$ ⇒ SSS (3). Perimeter 12 cm (2)	3+2
20	(i) 27 cm (ii) $38 - 27 = 11$ cm each (iii) $DR = 25 - 11 = 14$; ORDS is a square ⇒ $r = 14$ cm	1+2+2
	OR: RHS congruence ⇒ $PA = PB$ (3); $\triangle PAB$ isosceles ⇒ base angles equal (2)	3+2

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5 cm	Ratio $\frac{1}{2}$
2	**(c)** 6 cm	$\frac{3}{DB} = \frac{1}{2}$
3	**(c)** RHS	RHS is a congruence rule
4	**(b)** Correct	Ratio of sides 1, equal angles
5	**(a)** Student A is correct	DEF is the larger triangle
6	**(c)** 90°	Theorem 10.1
7	**(a)** 6 cm	$\sqrt{100-64}$
8	**(b)** 60°	Supplementary
9	**(d)** Incorrect — exactly one tangent can be drawn	Definition
10	**(a)** Student A is correct	$2\sqrt{100-64}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$DB = 4.5$; $\frac{3}{4.5} = \frac{AE}{3} \Rightarrow AE = 2$ cm	1+1
12	$\angle C = 65^\circ \Rightarrow \angle B = 180^\circ - 97^\circ = 83^\circ$	1+1
13	$\sqrt{1681-81} = 40$ cm	2
14	$2(4+3+5) = 24$ cm	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{1.5}{3} = \frac{1}{2} \Rightarrow DE \parallel BC$; $DE = \frac{1}{3} \times 6 = 2$ cm	2+1
16	Let M be the mid-point of arc AB. OM bisects $\angle AOB$, so $OM \perp AB$. Also $OM \perp$ tangent at M $\Rightarrow$ tangent $\parallel AB$	1+1+1
17	Scale factor $\frac{3}{2}$: $EF = 9$, $FD = 12$ cm; perimeters 18 : 27 = 2 : 3	2+1
	OR: $\triangle AOB \sim \triangle COD \Rightarrow \frac{OA}{OC} = \frac{OB}{OD} \Rightarrow OD = 4$ cm	1+1+1
18	$AF = x$: $(8-x) + (12-x) = 10 \Rightarrow x = 5$; $AF = 5$, $BD = 3$, $CE = 7$ cm	1+1+1
	OR: OT $\perp$ PQ at R: $PR = 2.4$, $OR = 1.8$; $\triangle TRP \sim \triangle PRO \Rightarrow TP = \frac{PR \cdot OP}{OR} = 4$ cm	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 20 m per cm (ii) 80 m and 100 m (iii) perimeter 240 m $\Rightarrow$ ₹12,000	1+2+2
	OR: Let AD meet PQ at E. $\triangle APE \sim \triangle ABD$ and $\triangle AQE \sim \triangle ACD \Rightarrow \frac{PE}{BD} = \frac{AE}{AD} = \frac{QE}{CD}$; $BD = CD \Rightarrow PE = QE$	2+2+1
20	(i) OACB is a square $\Rightarrow$ 1.5 cm each (ii) $1.5\sqrt{2} \approx 2.12$ cm (iii) tangents from an external point are equal	2+2+1
	OR: Shortest-distance proof (3); both tangents $\perp$ the same diameter $\Rightarrow$ parallel (2)	3+2

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